

# Hybrid SMART GRID

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This system integrates traditional AC grids with DC systems, such as solar panels, batteries, and EV chargers, and coordinates them through software, automation, and cloud platforms. In a conventional electrical grid, electricity flows from power generation stations to consumers, and when generation and consumption remain balanced, the grid operates efficiently.

When load (power demand) increases, the generator turbines experience additional mechanical stress, causing a reduction in their rotational speed. This decrease in speed results in a drop in frequency, which in turn leads to a decline in voltage. When voltage and frequency fall below normal thresholds, electrical equipment at the consumer end may malfunction. If this condition persists, the generated power becomes unstable and ineffective, and generators disconnect automatically to safeguard the system.

Preventing such instability requires reducing loads during these conditions. Traditionally, this reduction has been carried out manually at substations. Advancements in technology have enabled these operations to be automated, forming the basis of what is known as a smart grid.

In many countries, alternating current (AC) is distributed to consumers via transmission lines. With the growing adoption of DC-based devices, such as electric vehicles, and renewable energy sources, such as solar panels, integrating AC and DC systems has become increasingly important.

A system that combines AC and DC power networks is called a

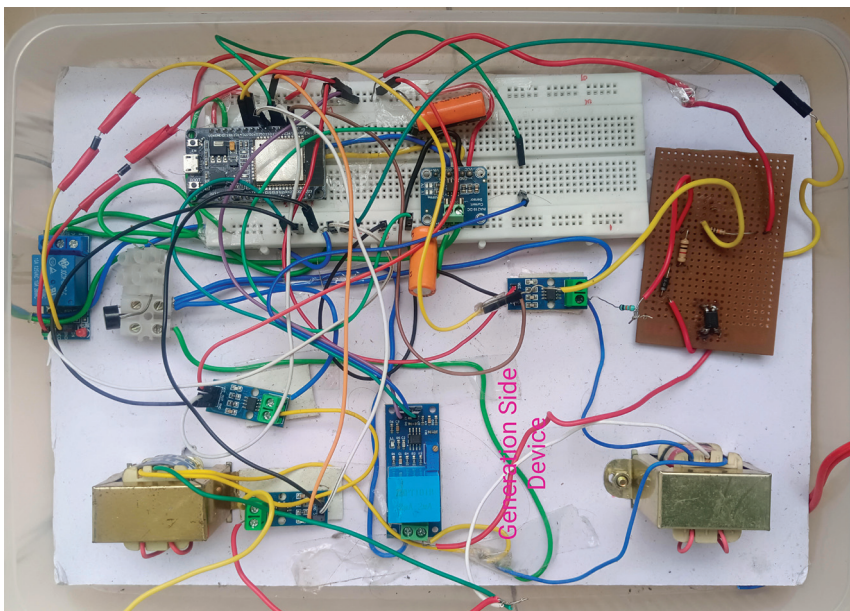


Fig. 1: Authors' generator-side prototype

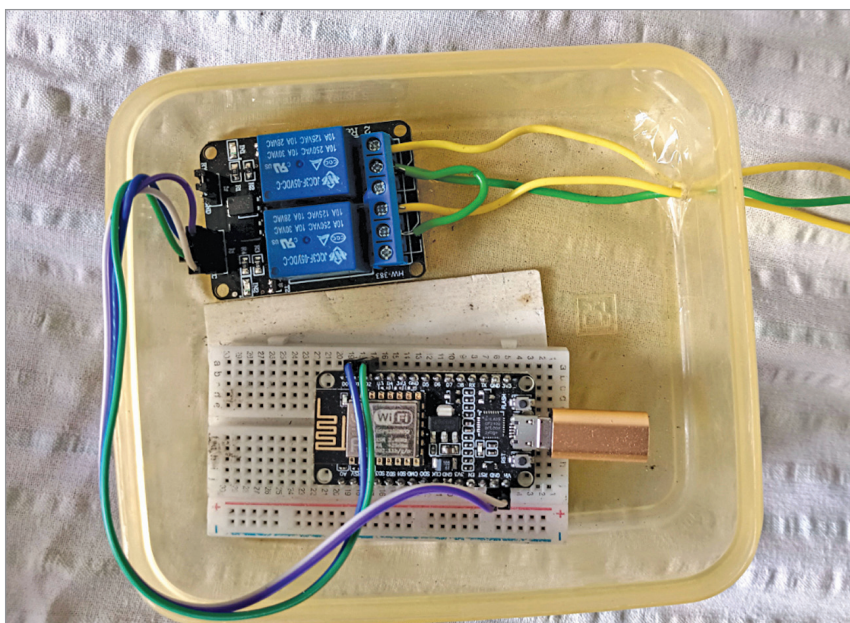


Fig. 2: Authors' load-side prototype

hybrid smart grid. These systems improve energy management efficiency and optimise power utilisation. This hybrid smart grid system is designed as a cost-effective,

educational platform that enables students, engineers, and designers to study and extend the concept for future applications.

The system includes two